

Other Models

Queueing System with Variable Service

Let us consider a queueing system where the service rate depends on the number of customers in the system. For example, a second server may be added when the queue length exceeds a critical value L . If arrivals follow a Poisson process and the service times are memoryless, then the transition rates of the birth-death process are

$$\begin{aligned}\lambda_n &= \lambda, \\ \mu_n &= \begin{cases} \mu, & n \leq L \\ 2\mu, & n > L. \end{cases}\end{aligned}$$

Other Models

Queueing System with Variable Service

In general: the arrival process is a Poisson process with rate $\lambda_n = \lambda$, and the service rates are arbitrary μ_n .

So, the stationary distribution in this case is

$$p_n = \frac{\lambda^n}{\mu_1 \mu_2 \dots \mu_n} p_0$$

where

$$p_0 = \left[1 + \sum_{k=1}^{\infty} \frac{\lambda^k}{\mu_1 \mu_2 \dots \mu_k} \right]$$

is the normalizing constant.

Other Models

Queueing System with Feedback

The process $\{N(t), t \geq 0\}$ is a birth and death process with

$$\lambda_n = \lambda, n \geq 0$$

$$\mu_n = q\mu, n \geq 1, q = 1 - p.$$

The stationary distribution $p_n = \lim_{t \rightarrow \infty} P(N(t) = n)$ exists if

$\frac{\lambda}{q\mu} < 1$. It is given by

$$p_n = \left(1 - \frac{\lambda}{q\mu}\right) \left(\frac{\lambda}{q\mu}\right)^n, n \geq 0.$$

Other Models

Queueing System with Discouragement

Let us consider the $M/M/s$ queueing system where the waiting time is limited to T_w (if the customer is not served before T_w expires, they leave the system without receiving service). Suppose that T_w is exponentially distributed with parameter ν . It is important to note that: if $\nu \rightarrow 0$, we have a system with rejected demands; if $\nu \rightarrow \infty$, we have a system with unlimited capacity. The discouragement can also be due to the length of the queue.

The state of the system at time t is always described by the process (3) with the state space

$$S = \{0, 1, 2, \dots, s, s+1, \dots, s+i\}, i > 1.$$

Other Models

Queueing System with discouragement

In the stationary regime, we have the following balance equations:

$$\left\{ \begin{array}{l} \lambda p_0 = \mu p_1 \\ \text{-----} \\ (\lambda + n\mu)p_n = \lambda p_{n-1} + (n+1)\mu p_{n+1}, 1 \leq n < s; \\ (\lambda + s\mu)p_s = \lambda p_{s-1} + (s\mu + \nu)p_{s+1}, n = s; \\ (\lambda + s\mu + i\nu)p_{s+i} = \lambda p_{s+i-1} + (s\mu + (i+1)\nu)p_{s+i+1} \\ , i \geq 1, n > s; \end{array} \right.$$

with $\sum_{n=0}^{\infty} p_n = 1.$

Other Models

Queueing System with discouragement

The solution of this system provides the stationary distribution

$$p_n = \frac{1}{n!} \left(\frac{\lambda}{\mu} \right) p_0, 1 \leq n < s;$$

$$p_{s+i} = \frac{\lambda^{s+i}}{s! \mu^s \prod_{n=1}^i (s\mu + n\nu)} p_0.$$

For the performance measures of this model, we have:

$$\overline{n_q} = \sum_{i=1}^{\infty} i p_{s+i}.$$

And the probability that a customer leaves without being served is:

$$p_{\text{leave}} = \frac{\nu \overline{n_q}}{\lambda}.$$

Other Models

Tandem Queueing System

We consider the following queueing system: a customer arrives according to a Poisson process $\lambda > 0$, and is first served by server A , then by server B . The service times at A and B are two independent random variables, each exponentially distributed with means $\frac{1}{\mu}$ and $\frac{1}{\nu}$, respectively. The service discipline is FCFS for both subsystems.

The state of the system at time t is described by a stochastic process $\{N_A(t), N_B(t)\}$, where $N_A(t)$ ($N_B(t)$) is the number of customers in subsystem A (subsystem B) at time t . This process is a homogeneous Markov chain with state space $S = \{(i, j), i \in \mathbb{N} \text{ and } j \in \mathbb{N}\}$.

Departures from the queueing system with Poisson arrivals and exponential service form a Poisson process.

Other Models

Tandem Queueing System

For the subsystem A :

$$p_n = \lim_{t \rightarrow \infty} P(N_A(t) = n) = \left(1 - \frac{\lambda}{\mu}\right) \left(\frac{\lambda}{\mu}\right)^n, n \geq 0.$$

For the subsystem B :

$$p_m = \lim_{t \rightarrow \infty} P(N_B(t) = m) = \left(1 - \frac{\lambda}{\nu}\right) \left(\frac{\lambda}{\nu}\right)^m, m \geq 0.$$

Consequently,

$$\begin{aligned} p_{nm} &= \lim_{t \rightarrow \infty} P(N_A(t) = n, N_B(t) = m) = p_n p_m \\ &= (1 - \rho_A) \rho_A^n (1 - \rho_B) \rho_B^m. \end{aligned}$$

Other Models

Tandem Queueing System

Regarding performance measures,

$$\overline{n}_A = \frac{\rho_A}{1 - \rho_A} \text{ and } \overline{n}_B = \frac{\rho_B}{1 - \rho_B}$$

and

$$\overline{n} = \overline{n}_A + \overline{n}_B.$$

Note: If the number of subsystems is greater than 2, the cascade is subdivided into as many subsystems as it contains, and each subsystem is studied separately.

Particular Queueing Systems (Markovian Models)

There are many Markovian queueing models for which the stochastic process (3) has a complex structure: models where arrivals (or service) occur in groups, models with classes of customers, etc. The process (3) associated with these models is no longer a birth-death process because transitions from a given state can occur to any other state.

Queueing System with batch arrivals

$M^X/M/1$ Model: Customers arrive in the system in groups according to a Poisson process ($\lambda > 0$). The number of customers per group is a strictly positive random variable X : $P(X(t) = x) = c_x(t)$. Service is provided by a single server. The service times follow a common distribution, which is exponential with a finite mean ($\frac{1}{\mu}$). Upon the arrival of a group, if the server is free, the first customer of that group is immediately attended to, while the other customers wait (the waiting capacity is unlimited). Otherwise, all customers in the group join the queue. The discipline is FCFS. We assume that all introduced variables (the time between consecutive arrivals, the service time, and the group size) are mutually independent.

Queueing System with batch arrivals

Stochastic Analysis

The state of the system at time t can be described by the stochastic process $\{N(t), t \geq 0\}$, where $N(t)$ is the number of customers in the system at time t . The state space is $S = \{0, 1, 2, \dots\}$.

Transitory state of the system:

let $p_n(t) = P(N(t) = n), n \geq 0$ and

$c_k(t) = P(X(t) = k), k \geq 1$. From the transition graph we get the state's probabilities equations.

Then the Kolmogorov equations system is :

$$\begin{cases} p'_0(t) = -\lambda p_0(t) + \mu p_1(t) \\ p'_n(t) = \lambda \sum_{k=1}^n p_{n-k}(t) c_k(t) - (\lambda + \mu) p_n(t) + \mu p_{n+1}(t) \\ \quad , n \geq 1. \end{cases}$$

Queueing System with batch arrivals

Stochastic Analysis

Stationnaire state of the system:

Let $p_n = \lim_{t \rightarrow \infty} p_n(t)$, $n \geq 0$ and
 $c_n = \lim_{t \rightarrow \infty} c_n(t)$, $n \geq 0$, ($c_0 = 0$).

The balance equations are

$$\begin{cases} \lambda p_0 = \mu p_1 \\ 0 = \lambda \sum_{k=1}^n p_{n-k} c_k - (\lambda + \mu) p_n + \mu p_{n+1}, n \geq 1. \end{cases} \quad (8)$$

Queueing System with batch arrivals

Stochastic Analysis

To solve the above system, we use the generating functions method. We introduce the following generating functions:

$$P(z) = \sum_{n=0}^{\infty} p_n z^n \text{ and } C(z) = \sum_{n=0}^{\infty} c_n z^n.$$

We then obtain the generating function for the stationary distribution of the number of customers in the system:

$$P(z) = \frac{\mu p_0 (1 - z)}{\mu (1 - z) - \lambda z [1 - C(z)]}, \text{ si } |z| < 1. \quad (9)$$

Queueing System with batch arrivals

Stochastic Analysis

To define ρ_0 , we use the normalisation equation $P(1) = 1$.
So, from (9) for $z \rightarrow 1$ we obtain

$$\rho_0 = 1 - \frac{\lambda E[X]}{\mu};$$

where $E[X] = C'(1) = \bar{c}$ and the total traffic intensity is $\rho = \frac{\lambda E[X]}{\mu}$. And the condition of the existence of the steady state is $\rho < 1$. That it gives

$$\rho_0 = 1 - \rho,$$

and

$$P(z) = \frac{\mu(1-z)(1-\rho)}{\mu(1-z) - \lambda z[1 - C(z)]}, \text{ if } |z| < 1. \quad (10)$$

Queueing System with batch arrivals

Performance Measures

To calculate the performance measures, we apply the generating function properties:

The average number of customers in the system

$$\bar{n} = P'(1) = \lim_{z \rightarrow 1} P'(z) = \frac{\rho + \frac{\lambda}{\mu} C''(1)}{2(1 - \rho)}.$$

Queueing System M/M/1 with Different classes of customers and absolute priority

In general, it is assumed that the customer population is homogeneous, meaning that the service times of the customers are identically distributed according to a common probability distribution. In more complex cases where customers are divided into classes, each class can be identified by its own service time distribution (heterogeneous population).

Queueing System M/M/1 with Different classes of customers and absolute priority

When there are priorities among clients, they can be relative or absolute

Relative priority: once service has started, a non-priority customer's service is not interrupted upon the arrival of a higher-priority customer this is "**non-preemptive priority**".

Absolute priority: the opposite case "**preemptive priority**"; in the latter case, the interrupted service will either resume where it left off the "**resume rule**" or start over completely the "**repeat rule**".

Queueing System M/M/1 with Different classes of customers and absolute priority

Model Description

Consider a Markovian M/M/1 queueing model with two classes of customers (1 and 2), with class 1 having absolute priority. The arrival rates are λ_1 and λ_2 for the two classes, respectively, and the service rates are μ_1 and μ_2 . Furthermore, for class 2, given that the service time distribution is exponential (memoryless property), resuming the service from where it was interrupted is equivalent to restarting it from the beginning.

Queueing System M/M/1 with Different classes of customers and absolute priority

Stochastic Analysis

The state of the system is described by the Markovian process $\{N(t) = N_1(t) + N_2(t), t \geq 0\}$, where $N_1(t)$ is the number of customers in class 1 and $N_2(t)$ the number of customers in class 2, at the instant t . Its state space is $S = \{(n_1, n_2) : n_1, n_2 = 0, 1, 2, \dots\}$.

Queueing System M/M/1 with Different classes of customers and absolute priority

Stochastic Analysis

Let $p_{n_1, n_2} = \lim_{t \rightarrow \infty} P(N_1(t) = n_1, N_2(t) = n_2)$; the equilibrium equations in steady state are:

$$\left\{ \begin{array}{l} \lambda p_{0,0} = \mu_1 p_{1,0} + \mu_2 p_{0,1}, n_1 = 0, n_2 = 0; \\ (\lambda + \mu_1) p_{n_1,0} = \lambda_1 p_{n_1-1,0} + \mu_1 p_{n_1+1,0}, n_1 > 0, n_2 = 0; \\ (\lambda + \mu_2) p_{0,n_2} = \mu_1 p_{1,n_2} + \lambda_2 p_{0,n_2-1} + \mu_2 p_{0,n_2+1}, \\ \quad n_1 = 0, n_2 > 0; \\ (\lambda + \mu_1) p_{n_1,n_2} = \mu_1 p_{n_1+1,n_2} + \lambda_2 p_{n_1,n_2-1} + \lambda_1 p_{n_1-1,n_2}, \\ \quad n_1 > 0, n_2 > 0. \end{array} \right.$$

Queueing System M/M/1 with Different classes of customers and absolute priority

Stochastic Analysis

The system can be solved using the generation function

$$F(z_1, z_2) = \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} p_{n_1, n_2} z_1^{n_1} z_2^{n_2}, \text{ such that } z_1, z_2 \in [0, 1].$$

After calculation and from normalization equation

$$F(1, 1) = 1, \text{ we obtain } p_{0,0} = 1 - \rho_1 - \rho_2, \text{ such that:}$$

$$\rho_1 = \frac{\lambda_1}{\mu_1}, \rho_2 = \frac{\lambda_2}{\mu_2}.$$

The stationary distribution exists if $p_{0,0} > 0$, so,

$$\rho = \rho_1 + \rho_2 < 1.$$

Queueing System M/M/1 with Different classes of customers and absolute priority

Stochastic Analysis

We then obtain

$$F(z_1, z_2) = \left(\frac{1 - \rho_1 - \rho_2}{1 - g(z_2) - \rho_2} \right) \left(\frac{1 - g(z_2)}{1 - z_1 g(z_2)} \right),$$

where

$$g(z_2) = \frac{1}{2\mu_1} \left[\mu_1 + \lambda_1 + \lambda_2 (1 - z_2) - \sqrt{(\mu_1 + \lambda_1 + \lambda_2 (1 - z_2))^2 - 4\lambda_1\mu_1} \right].$$

And the stationary distribution is also obtained:

$$p_{n_1, n_2} = \frac{1}{n_1! n_2!} \times \frac{\partial^{n_1+n_2} F(z_1, z_2)}{\partial z_1^{n_1} \partial z_2^{n_2}} \Big|_{z_1=z_2=0}.$$

Queueing System M/M/1 with Different classes of customers and absolute priority

Performance Measures

- The average number of class 1 customers in the system is

$$\overline{n}_1 = \frac{\partial F(z_1, z_2)}{\partial z_1} \Big|_{z_1=z_2=1} = \frac{\rho_1}{1 - \rho_1},$$

- The average number of class 2 customers in the system is

$$\overline{n}_2 = \frac{\partial F(z_1, z_2)}{\partial z_2} \Big|_{z_1=z_2=1} = \frac{\rho_2 + E[N_1]\rho_2}{1 - \rho_1 - \rho_2}.$$

- The average number of customers in the system is

$$\overline{n} = \overline{n}_1 + \overline{n}_2.$$

Networks Queues and Jackson Product Form

Introduction

Queueing networks expand the modeling potential of single queue systems, allowing for complex customer processes that require multiple, sequential treatments within interconnected queues. Customers can either proceed to other stations within the network or exit, with routing governed by deterministic or stochastic rules.

Queueing networks may involve multiple customer classes, each with distinct arrival, service, and priority protocols. This discussion focuses on single-class product-form networks, notable for their straightforward analytical solutions. Jackson networks, part of this category, have stationary product-form distributions and have been generalized through BCMP networks (named for Baskett, Chandy, Muntz, and Palacios), extending the applicability of this model type.

Networks Queues and Jackson Product Form

Open Jackson Networks

- An open Jackson network is a very general form of queueing network formed by interconnecting k " M/s " type stations.
- In a Jackson network, there are k servers, each with its own (unbounded) queue.
- Jobs at a server are served in FCFS order. The i^{th} server has service rate $\mathcal{E}(\mu_i)$.

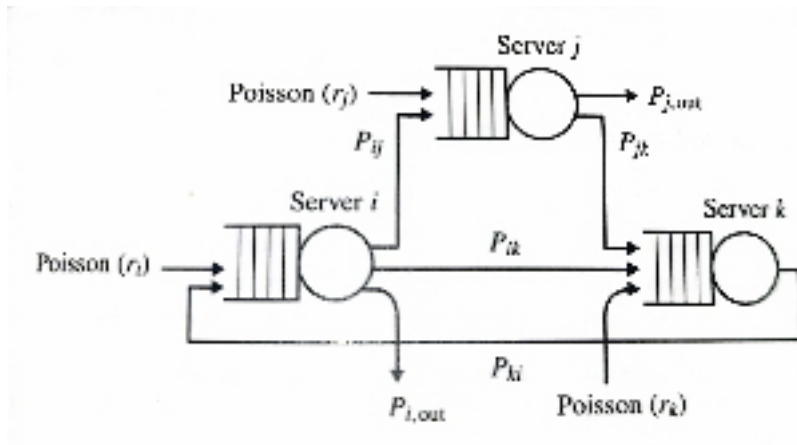
Networks Queues and Jackson Product Form

Open Jackson Networks

- Each server may receive arrivals from inside and outside the network. The arrivals to the network from outside constitute a Poisson process with rate λ .
- The routine of jobs is stochastic. Let P_{oi} be the probability that a customer arriving in the system goes to station i , and every job that completes at server i will be transferred to server j with probability P_{ij} , or will be exit the system with probability $P_{io} = 1 - \sum_{j=1}^k P_{ij}, j \neq i$.

Networks Queues and Jackson Product Form

Open Jackson Networks



Networks Queues and Jackson Product Form

Open Jackson Networks

The state of a Jackson network at time t is represented by the process $\{N(t) = (N_1(t), N_2(t), \dots, N_n(t)), t \geq 0\}$, where $N_i(t)$ is the number of customers in station i , $i = 1, \dots, k$. Given the assumptions about arrival and service processes and routing rules, it's straightforward to verify that $N(t)$ is a continuous-time Markov chain with a state space $S = \mathbb{Z}_+^n$. For a Jackson network to reach a stationary state with a steady-state distribution, the effective arrival rate at each station must be less than its maximum service rate.

Networks Queues and Jackson Product Form

Open Jackson Networks

Let λ_i denote the **effective arrival rate** at station i . In steady state, this arrival rate equals the departure rate and satisfies the **flow conservation** equations:

$$\lambda_i = \lambda e_i + \sum_{j=1}^k \lambda_j P_{ij}, \quad j \neq i$$

where e_i is the external arrival rate for node i . We can write

$$\lambda_i = r_i + \sum_{j=1}^k \lambda_j P_{ij}.$$

Networks Queues and Jackson Product Form

Open Jackson Networks

Let $N_1, N_2, \dots, N_k$ be the number of customers in the k nodes respectively as $t \rightarrow \infty$, and define

$$p_{n_1 n_2 \dots n_k} = P(N_1 = n_1, N_2 = n_2, \dots, N_k = n_k).$$

For the limiting distribution $p_{n_1 n_2 \dots n_k}$ Jackson has shown that

$$p_{n_1 n_2 \dots n_k} = p_1(n_1) p_2(n_2) \dots p_k(n_k), \quad (11)$$

where

$$p_i(n) = \begin{cases} p_i^{(0)} \frac{\left(\frac{\lambda_i}{\mu_i}\right)^n}{n!}, & 0 \leq n < s_i \\ p_i^{(0)} \frac{\left(\frac{\lambda_i}{\mu_i}\right)^n}{s_i^{n-s_i} s_i!}, & n \geq s_i \end{cases},$$

Networks Queues and Jackson Product Form

Open Jackson Networks

thus, if $\rho_i = \frac{\lambda_i}{\mu_i}$ is effective traffic intensity at each node, clearly $\rho_i < 1$ for $i = 1, 2, \dots, k$ for the limiting distribution to exist.

Remark: In steady state, a Jackson network behaves like isolated $M/M/s$ queueing systems, each receiving customers via independent Poisson processes. A distribution like (11) is called a **product-form** distribution, and networks with such a property are known as **product-form networks**.

Networks Queues and Jackson Product Form

Closed Jackson Networks

Jackson's initial work, which focused only on open networks, was extended and generalized multiple times, notably by Gordon and Newell, who demonstrated that product-form distributions also apply to closed networks composed of $-/M/c$ stations.

- Customers circulate between nodes according to transition probabilities p_{ij} , forming a closed system. In such networks, $e_i = 0$, $P_{io} = 0$, and $\lambda_i = \mu_i$ for all i ,
- the number of customers present in the system remains constant. If this number is equal to K , the system's state $N(t)$ at time t satisfies

$$K = \sum_{i=1}^k N_i(t) \text{ for all } t.$$

Networks Queues and Jackson Product Form

Closed Jackson Networks

A closed Jackson network is not only always stable but also has a stationary product-form distribution given by

$$p_{n_1 n_2 \dots n_k} = C \prod_{i=1}^k \frac{\rho_i^{n_i}}{a_i(n_i)}$$

where

$$a_i(n_i) = \begin{cases} n_i!, & 1 \leq n_i < s_i \\ s_i^{n-s_i} s_i!, & s_i \leq n_i \leq K \end{cases}$$

Networks Queues and Jackson Product Form

Closed Jackson Networks

Let the term $C = G(K)^{-1}$, it is a normalization constant ensuring that the sum of probabilities over all network states equals 1:

$$G(K) = \sum_{n_1 n_2 \dots n_k} \prod_{i=1}^k \frac{\rho_i^{n_i}}{a_i(n_i)}.$$

$$\rho_i = \frac{\lambda_i}{\mu_i} \text{ with } \lambda_i \text{ satisfying the relation } \lambda_i = \sum_{j=1}^k \lambda_j P_{ij}, j \neq i.$$